One Bit Off the Highway: Uniform Recovery for Langton's Ant

Hao Ke

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Abstract

We study a single colour perturbation applied at an arbitrary time and site along a canonical Langton highway, retaining its entire accumulated wake. An all-age coupling reduction expresses recovery through 104 guarded reference phases and finitely many younger exceptions. The resulting certificate specification has 739 cases, with a reported uniform bound of 64,584 updates and radius 78 after first impact. The same reduction yields an exact first-impact query and a compressed complete endpoint, with polynomial bit complexity in the time and perturbation coordinates. This manuscript develops the analytic reduction and specifies the finite certificate needed to establish the numerical recovery claims.

1 A growing highway and a single perturbation

The ant turns right on white and left on black, toggles the read cell, and moves one unit. Let F be this reversible update on a complete finite black field, position and heading. Start from the thirteen-black seed

$$P = \{(0,-2),(0,-1),(0,0),(1,-1),(2,-1),(2,1),(3,-2),\\ (3,1),(4,-2),(4,0),(5,-1),(5,0),(6,0)\}, \qquad X_0 = (P,(0,0),\mathrm{S}).$$

Its standard word W and drift $d = (2,-2)$ give $X_t = F^t X_0 = (B_t, p_t, h_t)$. We perturb one bit at an arbitrary site q:

$$Y(t,q) = (B_t \triangle \{q\}, p_t, h_t), \qquad t \geq 0,\ q \in \mathbb{Z}^2.$$

Thus an existing black cell can be removed as well as a white cell added. All other cells of the growing canonical history remain present.

The mathematical question is uniform recovery after the first physical encounter with the perturbation. Its waiting time can depend on an arbitrarily distant q. The reduction below separates this exact waiting-time query from the finite scattering problem and the all-age extension guard.

The following certificate target records the numerical statement of the research proof. The analytic implications are proved here; the release record tracks the corresponding finite data separately.

Theorem 1.1 (Certificate implication)**.** *Assume the finite records and guards specified in Section 4 pass their complete checks. Then every $Y(t,q)$ eventually follows standard P104. If $s \geq t$ is the first update at which the unperturbed orbit reads q, recovery is certified by time $s + 64584$. From impact to that boundary the perturbed head stays within Chebyshev distance 78 of q. If there is no such s, its entire future read trace agrees with the unperturbed one.*

Current-cell perturbations have at most 33,586 future intrinsic J_{10} events, attained at $t = 800$. Arbitrary-site perturbations have at most 33,595. At each fixed phase, all canonical ages of at least 25 periods have the same normalized infinite response to a current-cell flip.

An intrinsic J_{10} event tests the actual nine inverse turns and the current turn against the cyclic ten-factors of W. This definition retains the history across a perturbation, which may differ from the original canonical past.

2 The complete finite wake at arbitrary age

Let q_i be the pre-update positions of the 104 phases and T_r the XOR set of the first r read positions. The period write set is $T = T_{104}$. The finite period identity has deposited set

$$D = \{(0,-3),(0,0),(1,-4),(1,-3),(1,-2),(2,-1),\\ (2,0),(3,0),(6,-1),(7,-1),(7,0),(8,-1)\}.$$

Specifically, $F^{104}X_0$ has field $(P+d) \dot\cup D$ and pose (d, S), equivalently $D = P \triangle (P+d) \triangle T$. Distinct points of D lie in distinct full classes modulo $\mathbb{Z}d$, and $D - kd$ is disjoint from P for every $k \geq 1$. These are finite word and integer-ray checks.

Translate the state at time $104n$ by $-nd$. Its complete field is

$$C_n = P \dot\cup \bigcup_{k=1}^{n} (D - kd), \qquad |C_n| = 13 + 12n. \tag{1}$$

Telescoping the period write identity proves this formula by induction. At phase r the field is $C_n \triangle T_r$ and the pose is (q_r, h_r). For $n \geq N$, two such sources, with or without their current cell toggled, differ exactly on the finite set $\bigcup_{k=N+1}^{n}(D - kd)$.

3 Extending one recovery to every older source

Lemma 3.1 (First-difference coupling)**.** *If two complete fields at the same pose differ on E, and one trajectory never reads E, their entire read and pose traces agree; their field difference remains E throughout.*

Proof. Before a first differing read, both current colours agree. They turn together and toggle the same site, preserving the difference. Induction rules out such a first read. □

For an age N define the exclusion set $A_N = \bigcup_{k \geq N+1}(D - kd)$. It is a guard region; every physical source in (1) is finite.

Lemma 3.2 (Guarded age extension)**.** *Suppose the current-cell perturbation at age N, phase r, reaches a certified standard suffix. If its finite recovery footprint and all reads of that suffix avoid A_N, the same normalized infinite trace occurs at every age $n \geq N$. If its nine inverse reads also avoid A_N, all intrinsic event counts agree.*

Proof. The actual finite difference of the sources is contained in A_N. Apply Lemma 3.1 forwards and, for the history assertion, to the reversible rule for nine steps backwards. □

The infinite guard has a finite exact test. For a finite read p, solve $e - p = kd$ for each $e \in D$, with $k \geq N + 1$. For a suffix phase ray $v_i + je_{\text{out}}$, solve

$$v_i + je_{\text{out}} = e - kd, \qquad j \geq 0, \quad k \geq N + 1. \tag{2}$$

Nonparallel drifts give a unique rational solution by determinants, checked for integrality and the two inequalities. For equal parallel drifts and $v_i - e = ad$, a solution exists exactly when $-a \geq N+1$. For opposite drifts, equal full classes yield an eventual intersection. Distinct full classes never intersect. Thus every infinite ray and every old-wake epoch is covered.

4 The finite certificate specification

For each phase r, use the following age threshold N_r, in phase order:

```
8,7,9,2,1,5,1,3,5,15,7,2,2,1,2,6,1,2,5,2,3,11,21,3,4,3,
2,1,13,17,1,17,3,7,1,3,18,10,7,0,14,8,3,5,9,9,4,4,9,4,
5,25,9,13,2,12,7,2,0,16,9,14,12,14,0,3,2,4,14,3,4,8,11,
6,12,5,12,4,4,15,14,0,0,15,1,15,0,8,0,1,0,3,1,4,2,5,8,
1,3,0,3,2,1,1.
```

The certificate must contain exactly the pairs (n, r) with $0 \leq n \leq N_r$: 635 younger exceptions and 104 guarded bases. Each record specifies recovery length, endpoint pose, outgoing standard phase, exact intrinsic defect count and last defect time. Its checker must perform these steps:

1. Reconstruct the complete source (1), its phase writes and the current-cell flip; recover the actual nine inverse turns.
2. Replay every recovery update and compare the complete endpoint, pose, read word, event count and maximum distance from the impact cell.
3. Verify the entire proposed standard future by the parity test below, including its nine-letter incoming history.
4. At $n = N_r$, verify the finite, inverse and infinite footprints against A_{N_r} using (2).

At suffix phase i and epoch k, the necessary actual colour is

$$1_B(v_i + ke_{\text{out}}) \ \oplus \ \#\{(j, z) : v_j - v_i = ze_{\text{out}},\ 0 \leq z \leq k,\ z > 0 \text{ or } (z = 0, j < i)\} \pmod 2. \tag{3}$$

Here B is the entire endpoint field. A black cell hits a phase ray at most once, while each same-ray earlier phase supplies one threshold. Epoch zero, every hit and its successor, every self-write threshold, and the final infinite interval therefore form a finite complete check of (3). Induction over suffix time proves permanence.

The recorded aggregate targets are maximum recovery length 64,584 at $(n, r) = (22, 51)$, maximum radius 78 at $(25, 51)$, and maximum event count 33,586 at $(7, 72)$. The last corresponds to $t = 104 \cdot 7 + 72 = 800$. These are bounds at the certificate's recovery boundaries; the time bound is not a claim about the latest first permanent entry.

Current-cell part of Theorem 1.1. The explicit records cover $n < N_r$. Each guarded base covers all $n \geq N_r$ by Lemma 3.2. Taking the record maxima proves the bounds and the attained event count. Since $N_r \leq 25$, responses stabilize by age 25 at every phase. □

5 Arbitrary perturbation sites and exact endpoints

For arbitrary t, q, its first future canonical read is the minimum of

$$\{104n + i \geq t : q = q_i + nd,\ n \geq 0,\ 0 \leq i < 104\}.$$

Each phase supplies at most one n. Thus 104 exact collinearity and divisibility checks decide whether impact occurs and give its exact time. If there is none, coupling proves agreement forever. Otherwise the two orbits agree up to impact s, when the perturbed state is precisely X_s with its current cell toggled. The current-cell result now applies. Before impact all forward letters are canonical; only the first nine tests can see altered inverse history. Adding those nine to the post-impact bound gives 33,595 and completes Theorem 1.1.

Write $s = 104n + r$, choose $m = \min(n, N_r)$, and replay the bounded reference record at (m, r). If its normalized write set is Z and endpoint pose is $(p_{\mathrm{end}}, h_{\mathrm{end}})$, the true complete endpoint is

$$B_s \triangle \{q\} \triangle (Z + nd), \qquad (p_{\mathrm{end}} + nd, h_{\mathrm{end}}).$$

Formula (1) represents B_s by twelve finite arithmetic segments, the fixed seed and phase writes. Its binary size is polynomial in the input time and coordinates. The number of reference updates is bounded by an absolute constant. This proves the exact polynomial-bit-time endpoint algorithm, conditional on the same finite certificate.

6 Relation to existing results

The one-black theorem classifies complete inputs with one initially black cell. The present family instead contains $13 + 12n$ canonical historical cells before its phase writes, with one arbitrary colour toggle. The all-age guard is what extends a finite scattering check to those unbounded histories. The factor paper supplies the intrinsic event language; its local predicate is used here to count the recovery rather than to predict all possible initial fields.

A recovered standard trace can leave a finite scattering scar. Repeated perturbations therefore require an additional closure theorem for the complete repaired fields. The single-toggle family developed here supplies a concrete interface for that further problem.

The stable-read and symbolic-word viewpoints are prior methods [1, 2]. This manuscript's specific content is the full canonical-age quantifier, guarded recovery reduction and finite certificate specification. The repository review record identifies which data have been replayed and which must still be restored before numerical certification.

A Canonical word

The following concatenated lines specify the 104-letter word:

```
LRLLRRRRLLRLLRRRRLLRRRRLLRLRRRRLRLLLLRRRRLRRLRRRRLLLL
RLRRRRLRRRRLLLLRLRRRRLRLLRRLLLLRRLLRRRRLLRRLRLLRLLR.
```

The period and deposition identities can be checked from these letters and P directly. The required finite certificate and source manifest are identified in the accompanying evidence record.

References

[1] A. Gajardo. A symbolic projection of Langton's Ant. *Discrete Mathematics & Theoretical Computer Science*, 2003. doi:10.46298/dmtcs.2312.

[2] A. Jillhewar. *Finite-Support Periodic Highways of Langton's Ant: Necessary Conditions, Transverse Exclusions, and Exact Search*, 2026. Repository revision `76a7e234e577d992d086c204c3dea00ae72cf10a`, `https://github.com/Atharva12456/langtons-ant-highway`.